Linux Logging for SOC — Incident Investigation Report

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Investigation Type: Linux Log Analysis / Threat Hunting / DFIR

Platform: Linux

Lab: Linux Logging for SOC- TryHackMe

Date: September 2026

Executive Summary

This report documents the investigation of suspicious activity identified on a Linux system during the Linux Logging for SOC laboratory. The objective of the investigation was to analyze authentication logs, system logs, audit logs, and user activity to identify indicators of compromise, reconstruct attacker actions, and determine the overall security impact. The investigation identified evidence of repeated SSH authentication failures, successful login activity, suspicious command execution, privilege escalation behavior, and reconnaissance-related tooling usage. Correlation across multiple Linux log sources enabled reconstruction of the attacker timeline and identification of malicious behavior consistent with post-compromise activity.

1. Investigation Objectives

The primary objectives of the investigation were:

1. Detect unauthorized access attempts
2. Investigate successful authentication events
3. Identify suspicious process execution
4. Analyze audit logs for malicious activity
5. Detect privilege escalation behavior
6. Reconstruct attacker timeline

7. Identify indicators of compromise (IOCs)
8. Recommend mitigation and detection improvements

2. Log Sources Investigated

2.1 Authentication Logs

Authentication events were analyzed using:

/var/log/auth.log

Purpose

- SSH authentication monitoring
- Failed login detection
- Successful login analysis
- sudo activity monitoring
- User account modifications

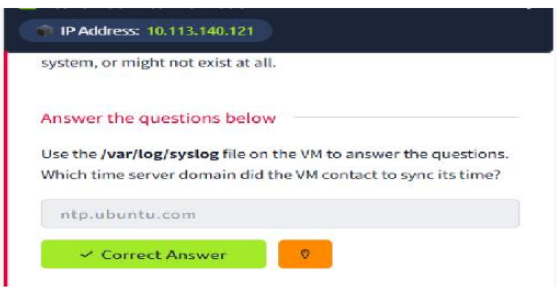
2.2 System Logs

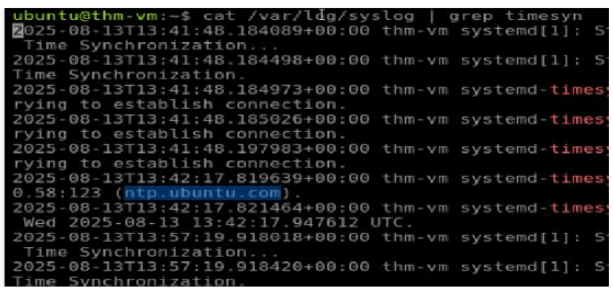
System activity was analyzed using:

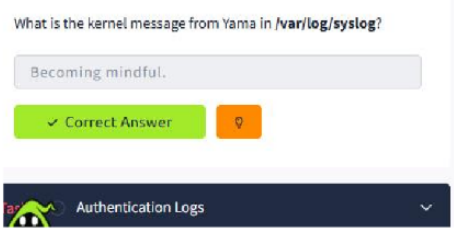
/var/log/syslog

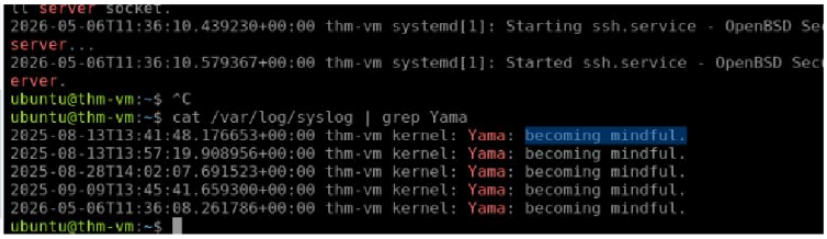
Purpose

- Service activity
- Process execution
- Scheduled tasks
- System alerts
- General operating system events









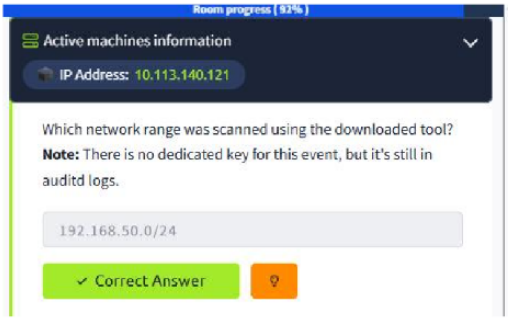
2.3 Audit Logs

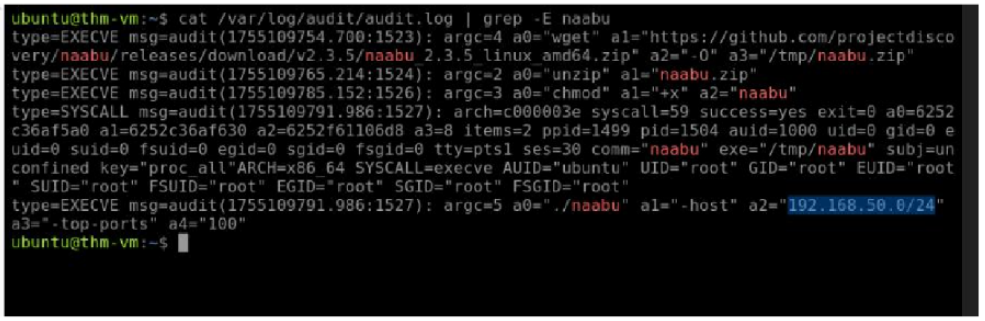
Detailed monitoring information was obtained from auditd:

/var/log/audit/audit.log

Purpose

- Process execution auditing
- File access tracking
- Command execution analysis
- User activity correlation





3. Investigation Methodology

The investigation followed a standard SOC triage and DFIR workflow:

1. Review authentication logs
2. Identify failed login attempts
3. Investigate successful authentication events
4. Correlate timestamps across logs
5. Analyze auditd process execution records
6. Review bash history artifacts
7. Reconstruct attacker timeline
8. Document findings and evidence

4. Authentication Analysis

4.1 Failed SSH Login Attempts

The following command was used:

```
grep "Failed password" /var/log/auth.log
```

Findings

Multiple failed SSH login attempts were identified from external IP addresses targeting multiple user accounts.

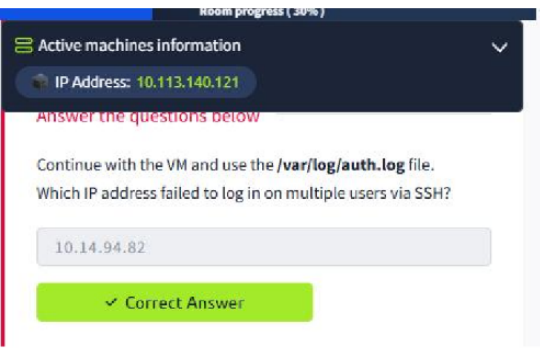
Analysis

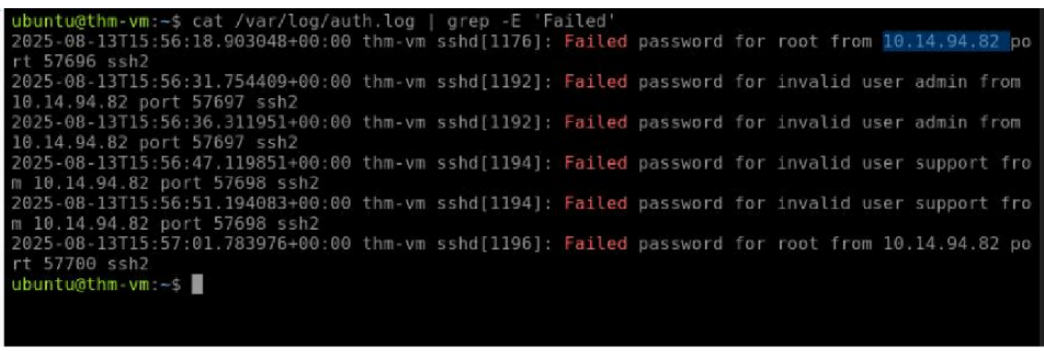
Observed behavior is consistent with:

- SSH brute-force attacks
- Password spraying
- Automated credential attacks

Security Impact

Repeated authentication failures indicate active adversary probing and unauthorized access attempts against the Linux host.





4.2 Successful Authentication Events

The following command was executed:

```
grep "Accepted password" /var/log/auth.log
```

Findings

Successful SSH authentication events were observed shortly after repeated failed login attempts.

Analysis

The sequence strongly suggests:

- Credential compromise
- Successful unauthorized access
- Potential attacker foothold establishment

Security Impact

Successful authentication following brute-force activity significantly increases the probability of system compromise.



4.3 sudo Activity Investigation

The following command was used:

`grep "sudo" /var/log/auth.log`

Findings

sudo command execution was identified following successful authentication activity.

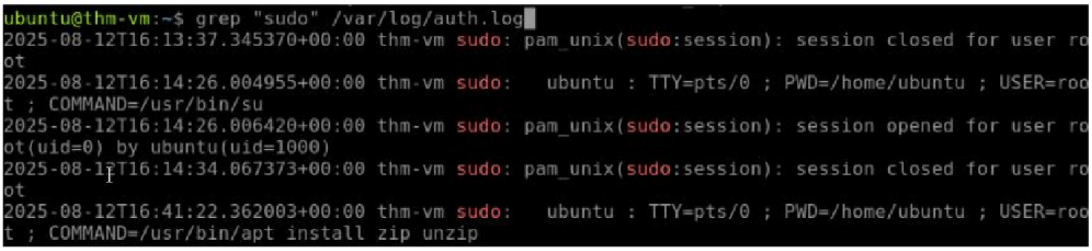
Analysis

This activity indicates potential:

- Privilege escalation
- Administrative access acquisition
- Post-compromise lateral activity

Security Impact

Unauthorized sudo execution may allow attackers to gain elevated privileges and perform administrative actions on the system.

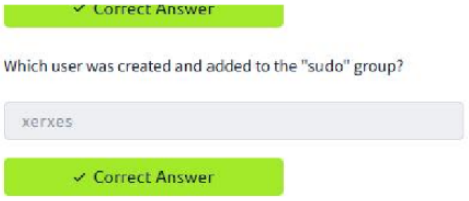


4.4 User Account Creation Activity

grep "useradd" /var/log/auth.log

Analysis

- New account creation activity was identified during the investigation timeline.
- Unauthorized account creation may indicate persistence establishment by an attacker.





5. Auditd Investigation

5.1 Process Execution Analysis

The following command was used:

ausearch -i

Additional filtering:

ausearch -k proc_wget

Findings

Suspicious process execution activity involving network utilities and file download tools was identified.

Observed commands included:

wget

curl

Analysis

This activity suggests:

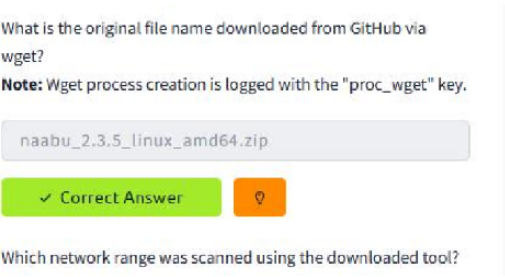
- External payload retrieval
- Tool downloading

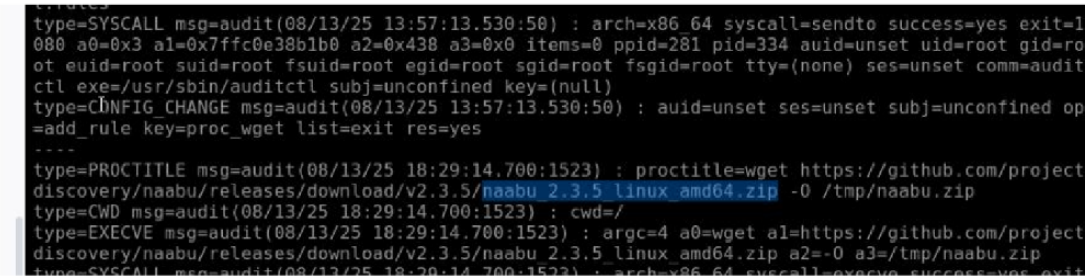
- Reconnaissance preparation
- Potential malware staging

Security Impact

Downloaded tooling may be used for persistence, reconnaissance, privilege escalation, or remote access operations.







5.2 File Access Monitoring

The following command was executed:

`ausearch -k file_thmsecret`

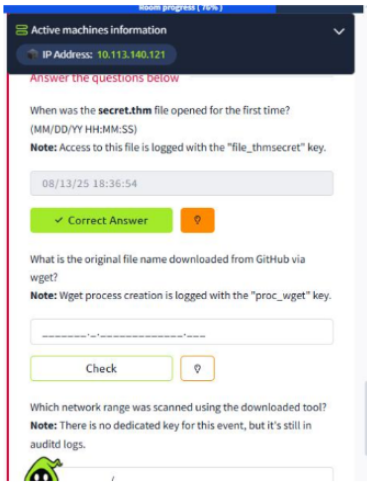
Findings

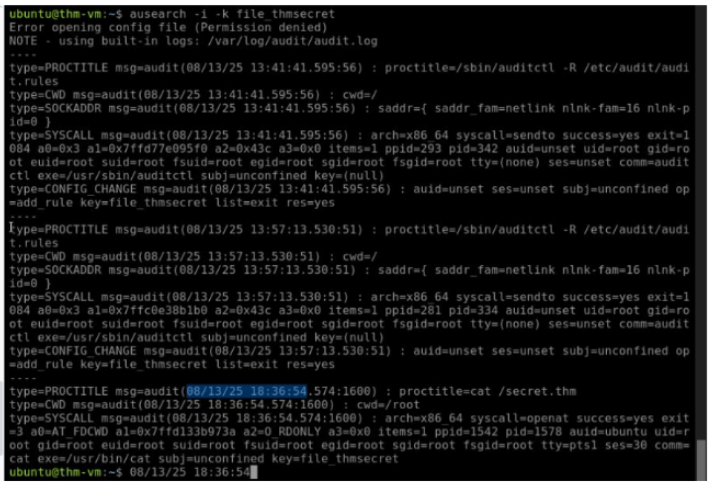
Sensitive file access events were identified during the investigation timeline.

Analysis

Observed behavior may indicate:

- Sensitive file enumeration
- Credential





discovery

- Data collection activity

Security Impact

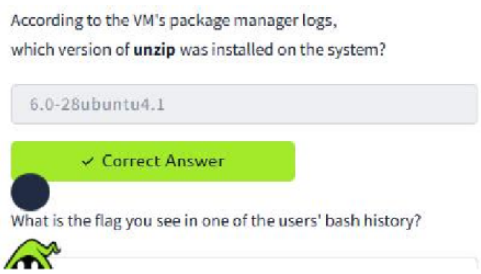
Unauthorized file access increases the risk of data exposure and credential compromise.

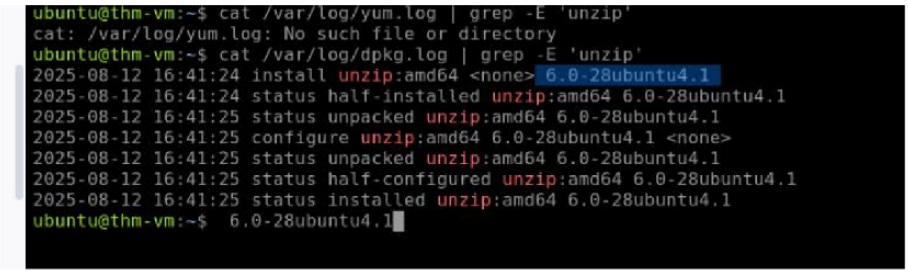
5.3 Package Installation Investigation

`grep -E "unzip" /var/log/yum.log`

Analysis

- Package installation logs revealed installation activity associated with archive extraction utilities.
- Such tools may be used during payload deployment or tool staging.





6. Bash History Analysis

The following artifact was investigated:

`~/.bash_history`

Findings

Command history revealed execution of multiple administrative and network-related commands.

Examples included:

`wget`

`curl`

`chmod`

sudo

useradd

Analysis

The commands indicate possible:

- Tool installation
- User modification
- Privilege escalation
- Persistence preparation
- Remote connectivity operations

Security Impact

Bash history artifacts provided direct evidence of potentially malicious command execution.



7. Timeline Reconstruction

Time	Event
T1	Multiple failed SSH login attempts detected
T2	Successful SSH authentication observed
T3	sudo activity initiated
T4	wget/curl commands executed
T5	Sensitive file access detected
T6	Reconnaissance-related activity observed

8. Indicators of Compromise (IOCs)

Network Indicators

Source IP ⟶ Suspicious external authentication source

Process Indicators

Wget ⟶ File download utility

Curl ⟶ Network communication utility

Sudo ⟶ Privilege escalation utility

9. Behavioral Indicators

- Repeated SSH authentication failures
- Successful login after brute-force activity
- Suspicious process execution
- Privilege escalation attempts
- Sensitive file access anomalies

10. Risk Assessment

Category	Severity
Unauthorized Access	High
Privilege Escalation	High
Reconnaissance Activity	Medium
Persistence Risk	Medium
Data Exposure	Medium

11. Recommendations

- Immediate Actions
- Disable compromised accounts

- Reset exposed credentials
- Block malicious IP addresses
- Review sudo permissions
- Isolate affected systems
- Conduct full malware scan

12. Long-Term Improvements

- Implement centralized logging
- Deploy SIEM monitoring
- Configure MFA for SSH access
- Harden SSH configuration
- Improve auditd coverage
- Monitor authentication anomalies
- Enable threat detection rules

13. Detection Opportunities

SSH Brute Force Detection

Monitor for:

Multiple failed SSH logins from same IP

Suspicious Process Execution

Monitor for:

`wget` OR `curl` executed by non-admin users

Privilege Escalation Detection

Monitor for:

Unexpected `sudo` activity

14. Conclusion

The investigation identified multiple indicators of suspicious and potentially malicious activity on the Linux host. Authentication analysis

revealed repeated SSH login failures followed by successful authentication events, suggesting possible credential compromise. Further analysis of audit logs and command execution records identified suspicious tooling usage, privilege escalation behavior, and sensitive file access activity. Correlation across Linux log sources enabled reconstruction of attacker actions and demonstrated the effectiveness of log analysis in SOC investigations.

The laboratory successfully simulated real-world SOC and DFIR investigative workflows involving Linux authentication monitoring, auditd analysis, timeline reconstruction, and threat hunting techniques.

15. Analyst Notes

This investigation demonstrated practical experience in:

- Linux log analysis
- SOC investigation methodology
- Threat hunting workflows
- Authentication monitoring
- auditd analysis
- Process execution investigation
- Timeline reconstruction
- Incident reporting

These skills align directly with responsibilities performed by SOC analysts, incident responders, and threat hunters in production security environments.